

MEMORY BASED QUESTION PAPER JEST-2019

Objective Questions:

Q.1 There are n students in a class. Class teacher sends some of the students to the drama section and rest to the music section. Teacher at the drama section selects some students for the play and assigns rest of the students backstage work. Teacher at the music section selects few students for vocals and assigns instruments to rest of the students. In how many ways work can be assigned? (Ans: 4^n)

Q.2 A, E, I, O, U and 15 Z's need to be arranged. Z can not occur at extremes. There should be at least 3 Z's between two vowels. How many such arrangements are possible? (Ans: $5! \cdot {}^6C_3 = 2400$)

Q.3 Let A be a 2×2 matrix. Trace of $A = 3$. $\text{Det}(A) = 3$. What is the trace of A^{-1} ? (Ans: 1)

Q.4 How many polynomials of degree ≥ 1 satisfy $f(x^2) = [f(x)]^2 = f(f(x))$? (Ans: 1)

Q.5 Solution in terms of asymptotic notation for $T(n) = 2T(n-2) + n$.

Q.6 An undirected graph is complete if there is an edge between every pair of vertices. Given a complete undirected graph on n vertices, in how many ways can you choose a direction for the edges so that there are no directed cycles? (TIFR 2017-B-12)

Q.7 Three dice are rolled independently. What is the probability that the highest and the lowest value differ by 4? (TIFR-2011-A-19)

Q.8 A random bit string of length n is constructed by tossing a fair coin n times and setting a bit to 0 or 1 depending on outcomes head and tail, respectively. The probability that two such randomly generated strings are not identical is? (Gate 2005-52)

Q.9 Let G be a simple undirected graph. Let TD be a depth first search tree of G . Let TB be a breadth first search tree of G . Consider the following statements.

- I. No edge of G is a cross edge with respect to TD . (A cross edge in G is between two nodes neither of which is an ancestor of the other in TD).
- II. For every edge (u, v) of G , if u is at depth i and v is at depth j in TB , then $|i - j| = 1$.

Which of the statements above must necessarily be true?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II (Gate 2018-30)

Q.10 Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either $j = i + 1$ or $j = 3i$. The minimum number of edges in a path in G from vertex 1 to vertex 100 is? (Gate 2005-IT-56)

Q.11 G is a simple undirected graph. Some vertices of G are of odd degree. Add a node v to G and make it adjacent to each odd degree vertex of G. The resultant graph is sure to be ? (Gate-2008-IT-27)

Q.12 Given a set of n distinct numbers, we would like to determine the smallest three numbers in this set using comparisons. Which of the following statements is TRUE?

- A. These three elements can be determined using $O(\log n)$ Comparisons.
- B. $O(\log n)$ comparisons do not suffice, however these three elements can be determined using $n+O(1)$ comparisons.
- C. $n+O(1)$ comparisons do not suffice, however these three elements can be determined using $n+O(\log n)$ comparisons.
- D. $n+O(\log n)$ comparisons do not suffice, however these three elements can be determined using $O(n)$ comparisons. (TIFR-2014-B-9)

Q.13 Let T be a depth first search tree in an undirected graph G. Vertices u and v are leaves of this tree T. The degrees of both u and v in G are at least 2. which one of the following statements is true?

- A. There must exist a vertex w adjacent to both u and v in G
- B. There must exist a vertex w whose removal disconnects u and v in G
- C. There must exist a cycle in G containing u and v
- D. There must exist a cycle in G containing u and all its neighbours in G (Gate 2006-48)

Q.14 In a complete k-ary tree, every internal node has exactly k children. The number of leaves in such a tree with n internal node is (Gate 2005-36)

Q.15 One question on regular expression (which was the only question asked from TOC).

Q.16 Happy numbers sad numbers.

Q.17 comparing cardinality of set of real numbers.

Q.18 hamming distance and set theory.

Subjective questions:

Q.1 How many functions from $\{1,2,3,\dots,n\}$ to $\{1995,1996\}$ such that $f(1) + f(2) + \dots + f(n)$ is odd. (Ans: 2^{n-1})

Q2. Given a sequence $a_1, a_2, a_3 \dots$ an of any different positive integers, exhibit an arrangement of integers between 1 and n^2 which has no increasing or decreasing subsequence of length $n+1$.

Q3. Prove that general solution of the equation $(x^2 + xy - y^2)^2 = 1$, is $\{f(n), f(n+1)\}$, with $n > 0$ where $f(n)$ is the term in Fibonacci sequence with $f(0) = 0, f(1) = 1$

Q4. Given a square matrix of size $n \times n$, elements are either 0 or 1. Suggest an algorithm to find the sub-rectangle made of 1, with largest area. (Find the area).

Q.5 Suppose that G contains a cycle C , and a path of length at least k between some two vertices of C . Show that G contains a cycle of length at least $\sqrt{k}$.

Question paper pattern:

Objective: 25 MCQ's most of them from previous TIFR, GATE Solving previous year questions from GO pdf is enough. Can expect more questions from graph theory, probability, counting). 3 marks for the correct answer -1 for incorrect.

Subjective: really challenging. 8 marks and 12 marks questions are asked. Most of them were about proofs.